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Project	Predict & Explain: Identifying Heart Disease Risk Factors
Subject	Data Science Mini Project – 2025–2026

Abstract

Heart disease remains one of the leading causes of death globally, yet a significant proportion of cases are preventable through early detection and timely intervention. This mini project focuses on building a machine learning-based classification system that predicts the likelihood of heart disease in a patient using clinical diagnostic features such as age, sex, chest pain type, resting blood pressure, serum cholesterol, fasting blood sugar, maximum heart rate achieved, and ST depression during exercise.

The dataset used is the Cleveland Heart Disease dataset comprising 303 patient records with 13 clinical attributes and a binary target variable indicating the presence or absence of heart disease. The pipeline begins with data preprocessing — handling missing values through median imputation and standardising features using StandardScaler — followed by Exploratory Data Analysis to understand distributions, correlations, and class balance.

Five classification models are trained and evaluated: Logistic Regression, Decision Tree, Random Forest, Gradient Boosting, and K-Nearest Neighbours. Each model is assessed using Accuracy, Precision, Recall, F1-Score, and AUC-ROC metrics. The K-Nearest Neighbours model achieved the best F1-Score of 69.57%, demonstrating that proximity-based similarity captures meaningful clinical patterns for heart disease classification.

Key findings from the analysis reveal that chest pain type, maximum heart rate, and ST depression (oldpeak) are among the strongest indicators of heart disease. The project aligns with the United Nations Sustainable Development Goal 3 — Good Health and Well-being, specifically Target 3.4, which aims to reduce mortality from non-communicable diseases by one-third by 2030.

Keywords: Heart Disease Prediction, Classification, Machine Learning, Clinical Data, Logistic Regression, Random Forest, KNN, SDG Goal 3